

Area and Perimeter (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
1(a)	3	3	M2 for $8x = 6(x+1)$ or M1 for Area of trapezium is $6(x+1)$
1(b)	22	1	FT 2(8 + their x)
2(a)	36	1	
2(b)	50	3	M2 for $8 \times 4 + 6 \times 3$ or $10 \times 3 + 4 \times 5$
			M1 for establishing area of one correct rectangular component
3(a)	$(2x-1)(x+2) = 18$	1	M1 for formulating the initial area product
	$2x^2 + 4x - x - 2 = 18$	1	M1 for correct expansion leading to given equation
3(b)	$\frac{5}{2}$	3	M2 for $(2x-5)(x+4) = 0$ or correct substitution into quadratic formula
			M1 for identifying $2x^2 + 8x - 5x - 20 = 0$
3(c)	17	2	M1 for substituting $x = 2.5$ into $2(2x-1+x+2)$
4(a)	(1,6)	1	B1 for correct coordinates.
4(b)	16	2	M1 for base = 4 and perpendicular height = 4. Calculation 4×4 .
4(c)	$2\sqrt{13}$	3	M1 for using distance formula $\sqrt{(5-(-1))^2 + (6-2)^2}$.
			M1 for simplifying to $\sqrt{36+16} = \sqrt{52}$. A1 for final surd form.
5(a)	55	2	M1 for $\frac{1}{2} \times (8+14) \times 5$
5(b)	$\frac{1}{2} \times 11 \times h = 55$	1	M1 for correctly equating area expression to 55

Part	Answer	Marks	Partial Marks / Guidance
	$h = 10$	1	A1 for clean algebraic isolation without errors
6(a)	30	3	M1 for $AC^2 = 5^2 + 12^2 = 169$. M1 for $AC = 13$. A1 for $5 + 12 + 13 = 30$.
6(b)(i)	3 hours 0 minutes	2	M1 for $\text{Time} = \frac{30}{10} = 3$. A1 for explicit conversion to 3 hours 0 minutes.
6(b)(ii)	36	3	M1 for area scale factor = 1.44 or $\frac{144}{100}$.
			M1 for length scale factor = $\sqrt{1.44} = 1.2$. A1 for $30 \times 1.2 = 36$.
7(a)	$\text{Area} = (2x + 1)(x + 2)$	M1	For correctly setting up the area product.
	$2x^2 + 4x + x + 2 = 27$	M1	For correct expansion of the brackets.
	$2x^2 + 5x - 25 = 0$	A1	For correctly rearranging to form the required equation with no errors seen.
7(b)	$x = 2.5$ or $x = \frac{5}{2}$	2	M1 for $(2x - 5)(x + 5) = 0$ or correct use of formula. Discards $x = -5$.
7(c)	21	2	M1 for substituting $x = 2.5$ into length and width to get 6 and 4.5, then adding and multiplying by 2.
8(a)	‘9’ and ‘15’	3	M1 for setting up equation ‘ $\frac{1}{2}(3k + 5k) \times 8 = 96$ ’
			M1 for simplifying to ‘ $32k = 96$ ’ or ‘ $k = 3$ ’
8(b)	‘ $24 + 2\sqrt{73}$ ’	4	M1 for identifying the base of the right-angled side triangle is ‘ $\frac{15-9}{2}$ ’ or ‘3’
			M1 for formulating Pythagorean theorem ‘ $\sqrt{3^2 + 8^2}$ ’
			M1 for exact slant side ‘ $\sqrt{73}$ ’